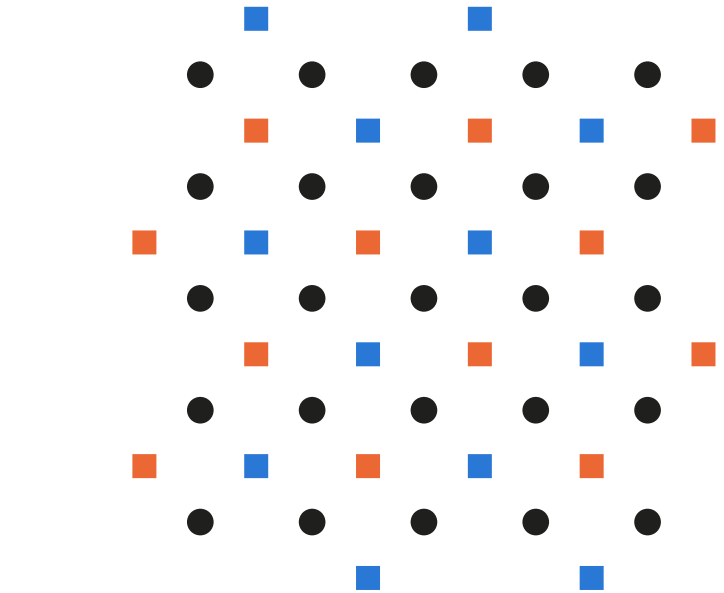
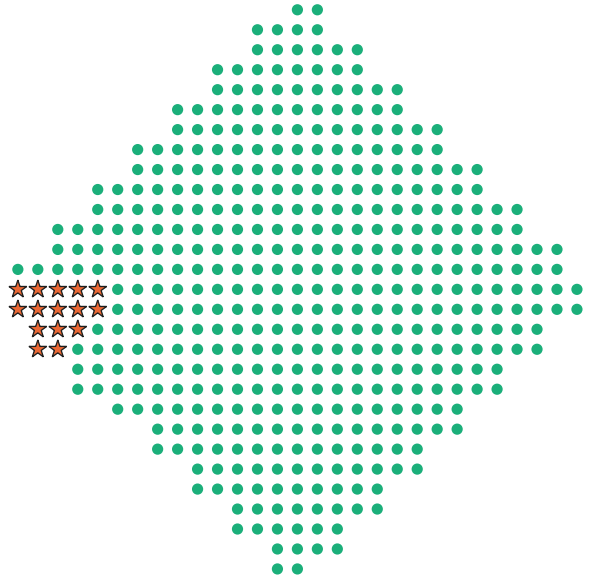
F7 · Physical circuits simulated with Stim (uniform depolarizing p = 0.1%)

(a) Data memory: surface code d = 5, 49 qubits
5 QEC cycles per shot · 120 detectors · decoded by PyMatching
p = 0.1% → 1 in 38,000 logical error per cycle



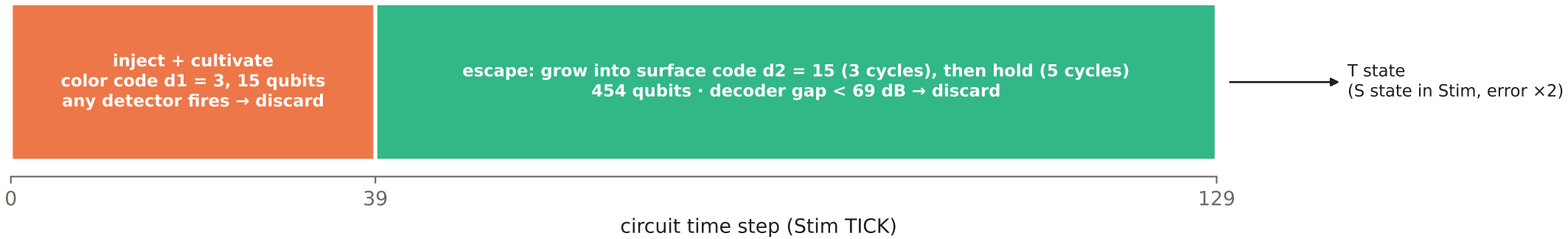
data qubits (25) X check qubits (12) Z check qubits (12)

(b) Magic-state factory (end-to-end): 454 qubits, 2,162 detectors
p = 0.1% → 1 in 479,000 error per T state, 77.5% discarded



grown surface code d2 = 15 (439 qubits)
injection + cultivation, color code d1 = 3 (15 qubits)

(c) One factory shot over time: 129 time steps



Layouts from the circuits' own QUBIT_COORDS. (a) stim.Circuit.generated('surface_code:rotated_memory_z', distance=5, rounds=5). (b, c) cultiv.make_end2end_cultivation_circuit(dcolor=3, dsurface=15, basis='Y', r_growing=3, r_end=5, inject_style='unitary'); stars = qubits used in the first 39 time steps (= length of the injection + cultivation circuit).